Module 2 Activity 2
Transient Response of Sensors

Objectives

1. Acquire time-varying data using Arduino and the Adafruit Data Logger Shield
2. Measure the time constant of a light-dependent resistor
3. Measure the time constant of an Arduino joystick module

Materials

1. Arduino
2. Adafruit Data Logger Shield
3. Light-dependent resistor
4. LED
5. Arduino joystick module
6. Data Streamer

Transient and Steady State Response

A control system's transient response is the temporary behavior exhibited by the system as it transitions from its initial state to a steady-state condition following a disturbance or input change. This transient behavior is characterized by a dynamic response that may involve overshoot, undershoot, ringing, or other oscillatory phenomena before settling to a new equilibrium point - the steady state [1].

Time constant

The time constant describes the speed of the response of a system to an input. For a first-order control system responding to a step or impulse input, the time constant is the time it takes for the system to reach 63.2% of the total rise (or decay) [2]. Figure 1 shows an example of a step response of a first-order system and its corresponding time constant.

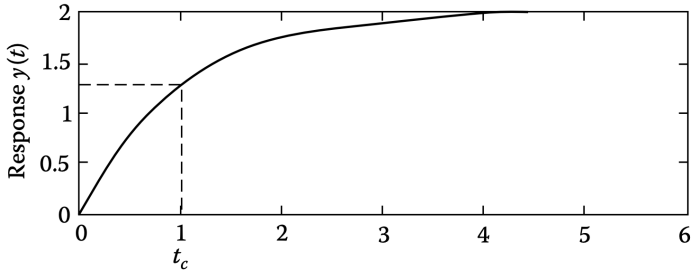


Figure 1. Step response of a first-order system and its corresponding time constant [2]

Light-dependent resistor

An Arduino UNO microcontroller can be used to interface with a light-dependent resistor (LDR) to measure ambient light levels. Figure 2 shows the connection of the LDR and Arduino. The LDR, a type of resistor that changes its resistance based on the intensity of light, is connected to the Arduino's analog input pins. The Arduino then reads the analog voltage corresponding to the LDR's resistance [3,4].

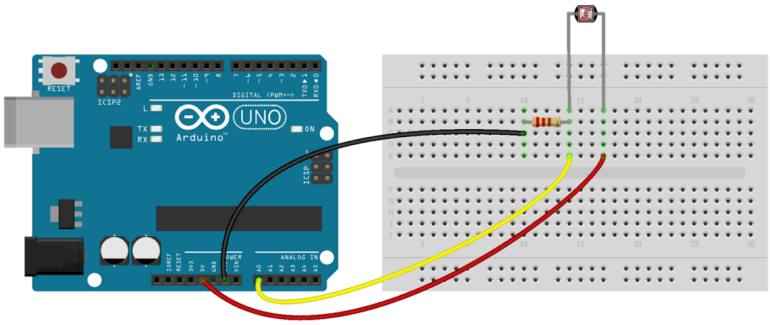


Figure 2. LDR light sensor circuit [3]

By analyzing the changes in resistance, the Arduino can determine whether the environment is dark, dim, light, or bright. This simple setup provides a cost-effective and versatile solution for various applications, such as automatic lighting control or environmental monitoring [3,4].

Joystick Module

Connecting a joystick module to an Arduino UNO allows you to capture user input through directional movements and button presses. These modules consist of two internal potentiometers mounted at right angles and an integrated push button. They measure the horizontal position (X-coordinate), vertical position (Y-coordinate), and button press state, outputting continuous analog signals alongside a digital signal. The data collected can be used in many applications, such as controlling game characters, directing robotic vehicles, operating pan-tilt camera platforms, or navigating graphical menus on displays [5].

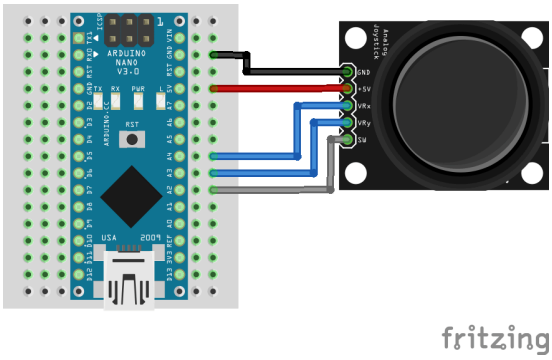


Figure 3. 5-pin joystick module [6].

Procedure

1. Using the Arduino and Adafruit Data Logger Shield, collect the time-varying output voltage from the light-dependent resistor, and the time-varying horizontal and/or vertical position of the joystick.
2. Calculate the rising time constant (dark to bright) and falling time constant (bright to dark) for a light-dependent resistor.
3. Calculate the rising time constant (neutral to far left/right/up/down) and falling time constant (snap-back to neutral) of the joystick.

Proposed Outline for Laboratory Report

1. Light-dependent resistor - construction and characteristics
2. Joystick module - construction and characteristics
3. Time constant - concept and usage
4. Circuit and code for time-varying voltage from the LDR, position from the joystick, and Arduino
5. Experiment to measure the time constant of sensors
    a. Procedure and results for LDR
    b. Procedure and results for joystick
6. Reflection and self-grade

References

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6. Arduino Project Hub (n.d.). Using joystick module with arduino. https://projecthub.arduino.cc/hibit/using-joystick-module-with-arduino-0ffdd4

Supplementary Materials

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2. *What is Data Streamer? - Microsoft Support*. (n.d.). https://support.microsoft.com/en-us/office/what-is-data-streamer-1d52ffce-261c-4d7b-8017-89e8ee2b806f
3. Adafruit Data Logger Shield. https://learn.adafruit.com/adafruit-data-logger-shield/overview